Phase 1 Technical Report: Problem Formulation, Data Understanding, and Quality Audit

NYC Yellow Taxi Hourly Demand Forecasting & Congestion Pricing Impact Analysis (2025)

Data Analysis Course Project (Fall 2026 / 1405)
Track A: NYC TLC Structured Time-Series Analysis

October 27, 2026

Abstract

This report presents Phase 1 of the NYC Yellow Taxi demand forecasting project for the full calendar year 2025. Leveraging a local DuckDB analytical engine operating directly over 12 monthly Parquet files (48.7 million trip records), we execute a comprehensive data structure and data quality audit. We formalize the operational decision problem—assisting fleet operators in pre-positioning vehicles across 265 taxi zones and evaluating the newly enacted 2025 Central Business District (CBD) congestion fee—by establishing the spatio-temporal aggregation unit $(z, t) \in \text{Zone} \times \text{Hour}$. Our audit reveals complete temporal continuity with zero null value records across core fields, while quantifying specific physical and financial anomalies: 546,304 non-positive duration trips (1.12%), 1,402,958 zero-distance trips (2.88%), and 2,848,620 negative fare entries (5.85%). Finally, we define a strict temporal split protocol (Train: Jan–Sep 2025; Test: Oct–Dec 2025) to prevent future data leakage and outline baseline models for subsequent implementation phases.

1 Problem Definition and Motivation

Taxicab demand in dense urban environments exhibits high spatial variance and non-stationary temporal dynamics. Following the 2025 implementation of the New York City Central Business District (CBD) congestion pricing fee (`cbd_congestion_fee`), fleet managers and independent operators face increased operational overhead and altered passenger booking behavior.

The core motivation of this project is to assist vehicle fleet managers in dynamically pre-positioning taxicabs across official NYC Taxi and Limousine Commission (TLC) zones. By accurately forecasting hourly trip originations per zone, dispatchers can minimize idle cruising time, reduce passenger wait times, and optimize fleet allocation under new congestion fee structures.

2 Related Work

Short-term urban passenger demand forecasting traditionally relies on historical rolling averages and Auto-Regressive Integrated Moving Average (ARIMA) models. Recent advances employ non-linear gradient-boosted decision trees (e.g., LightGBM, XGBoost) and spatio-temporal graph neural networks. In the context of NYC TLC datasets, prior literature emphasizes the necessity of strictly preserving temporal order during model validation to avoid look-ahead bias and spurious performance metrics.

3 Data and Data Quality

3.1 Dataset Overview

The primary dataset consists of the official NYC Yellow Taxi trip records for the entire calendar year 2025 (January 1, 2025 – December 31, 2025), partitioned into 12 monthly Parquet files stored locally

in `data/raw/`. Spatial reference lookups (`taxi_zone_lookup.csv`) and zone boundary shapefiles (`taxi_zones.shp`) are stored in `data/spatial/`.

3.2 Structural Audit

Using an in-memory DuckDB query engine, we processed all 12 Parquet files in a single unified operation without memory overflow. Table 1 lists the complete schema of the 2025 dataset, comprising 20 columns.

Table 1: NYC Yellow Taxi 2025 Dataset Schema and Variable Categorization

Column Name	Data Type	Category	Description / Role
`VendorID`	INTEGER	Categorical ID	TPEP provider code
`tpep_pickup_datetime`	TIMESTAMP	Datetime Variable	Trip start time (Meter engaged)
`tpep_dropoff_datetime`	TIMESTAMP	Datetime Variable	Trip end time (Meter disengaged)
`passenger_count`	BIGINT	Numeric	Driver-entered passenger count
`trip_distance`	DOUBLE	Continuous Numeric	Elapsed trip distance (miles)
`RatecodeID`	BIGINT	Categorical ID	Final rate code in effect
`store_and_fwd_flag`	VARCHAR	Flag	Memory buffer transmission flag
`PULocationID`	INTEGER	Spatial ID	Pickup TLC Taxi Zone ID (1..263)
`DOLocationID`	INTEGER	Spatial ID	Dropoff TLC Taxi Zone ID (1..263)
`payment_type`	BIGINT	Categorical	Payment method (1=Credit, 2=Cash)
`fare_amount`	DOUBLE	Financial Numeric	Metered time-and-distance fare
`extra`	DOUBLE	Financial Numeric	Miscellaneous extras and surcharges
`mta_tax`	DOUBLE	Financial Numeric	$0.50 MTA state tax
`tip_amount`	DOUBLE	Financial Numeric	Credit card tip amount
`tolls_amount`	DOUBLE	Financial Numeric	Total bridge/tunnel tolls
`improvement_surcharge`	DOUBLE	Financial Numeric	$0.30 improvement surcharge
`total_amount`	DOUBLE	Financial Numeric	Total charge to passenger
`congestion_surcharge`	DOUBLE	Financial Numeric	Standard congestion fee
`Airport_fee`	DOUBLE	Financial Numeric	Airport pickup fee
`cbd_congestion_fee`	DOUBLE	Financial Numeric	**New 2025 CBD Congestion Pricing Fee**

3.3 Spatial Data Structure and Schema Audit

To support spatial aggregations, zone-level demand forecasting, and geographic visualizations, we conducted a structural audit of the spatial reference files stored in `data/spatial/`.

3.4 Taxi Zone Lookup Table (`taxi_zone_lookup.csv`)

The lookup table maps numerical taxi zone identifiers to descriptive geographic attributes, comprising 265 total records and 4 columns, as detailed in Table 2.

Table 2: Taxi Zone Lookup Table Schema (`taxi_zone_lookup.csv`)

Column Name	Data Type	Category	Description / Role
`LocationID`	INTEGER	Primary Identifier	Unique zone identifier (1..265)
`Borough`	VARCHAR	Categorical	NYC Borough name (Manhattan, Queens, etc.)
`Zone`	VARCHAR	Descriptive	Neighborhood or specific zone name
`service_zone`	VARCHAR	Categorical	Service zone classification (Yellow Zone, Boro Zone, etc.)

3.5 Taxi Zones Spatial File (`taxi_zones.shp`)

The spatial vector dataset provides polygon boundaries corresponding to the taxi zones (stored as an ESRI Shapefile bundle with CRS EPSG:2263), comprising 263 distinct polygon records and 7 attributes, as listed in Table 3.

Table 3: Taxi Zones Spatial Vector Schema (`taxi_zones.shp`)

Column Name	Data Type	Category	Description / Role
`OBJECTID`	INTEGER	Identifier	ESRI object feature identifier
`Shape_Leng`	DOUBLE	Numeric Metric	Perimeter length of the zone polygon
`Shape_Area`	DOUBLE	Numeric Metric	Area of the zone polygon
`zone`	VARCHAR	Descriptive	Neighborhood zone name matching lookup table
`LocationID`	INTEGER	Spatial ID / Foreign Key	TLC Taxi Zone ID linking to lookup and trips
`borough`	VARCHAR	Categorical	Borough name classification
`geometry`	GEOMETRY	Spatial Data Type	Polygon / MultiPolygon boundary coordinates (EPSG:2263)

3.6 Table and Spatial Relationships

The trip records link dynamically to the spatial reference files using `PULocationID` and `DOLocationID` as foreign keys that join against the primary key `LocationID` in both the lookup table and the spatial geometry shapefile.

3.7 Data Quality Quantification and Decisions

The audit evaluated $48,722,602$ raw trip records. Table 4 details the findings for missing values, impossible physical measurements, and administrative anomalies, along with the corresponding handling strategy.

3.7.1 Rationale for Data Quality Treatment Decisions

To maintain data integrity and prevent distortion in our spatio-temporal demand forecasting models, we established explicit treatment protocols for each identified anomaly category across the $48,722,602$ raw records. The individual rationales and references to representative sample records are structured as follows:

- **Missing Timestamps and Spatial Location IDs (0 records, 0.00%):** *Decision: Retain All. Rationale:* The dataset exhibited complete structural integrity for core operational fields. Because

Table 4: Data Quality Audit Metrics and Treatment Decisions (Full Year 2025)

Metric / Anomaly Category	Count	% of Total	Initial Treatment Decision
Total Raw Trips	48,722,602	100.00%	Baseline dataset population
Missing Pickup Timestamps	0	0.00%	None required (100% complete)
Missing Dropoff Timestamps	0	0.00%	None required (100% complete)
Unknown Pickup Location (`PULocationID = 264`)	80,873	0.17%	**Filter Out**: Unmapped geographic pickup coordinates
Outside NYC Pickup Location (`PULocationID = 265`)	22,170	0.05%	**Filter Out**: Excluded for internal origin demand modeling
Unknown Dropoff Location (`DOLocationID = 264`)	105,193	0.22%	**Filter Out**: Unmapped geographic dropoff coordinates
Outside NYC Dropoff Location (`DOLocationID = 265`)	219,073	4.49%	**Conditional Retention**: Retained up to 200 miles for out-of-city flow metrics
Missing Distance / Fare Fields	0	0.00%	None required (100% complete)
Invalid Duration ($\leq$ 0 min or $>$ 240 min)	563,024	1.16%	**Filter Out**: Clock reset, meter left running, or cancelled trips
Invalid Trip Distance (Intra $>$ 100mi or Out $>$ 200mi)	2,429	0.005%	**Filter Out**: Extreme spatial anomalies and GPS satellite jumps
Zero Trip Distance ($d = 0$ mi)	1,402,958	2.88%	**Filter Out**: Stationary meter runs / cancellations
Invalid Average Speed ($v > 80$ mph)	560,004	1.15%	**Filter Out**: Telemetry errors and GPS coordinate displacement
Micro-Trip Speed Artifacts ($\Delta t < 1$ min, $v > 70$ mph)	8,543	0.018%	**Filter/Nullify**: Timestamp jitter and GPS rounding glitches
Negative Fare Amount ($f < \$0$)	2,848,620	5.85%	**Filter Out**: Dispute voids, chargebacks, system tests
Invalid Passenger Count ($<$ 0 or $>$ 9 or `null`)	11,611,894	23.83%	**Retain As-Is**: Retained without imputation to prevent data loss.

our fundamental unit of analysis is defined by pickup location and timestamp (z, t), having complete coverage in these fields ensures that no records are dropped due to missing spatial keys, eliminating potential sample selection bias at the initial ingestion stage.

- **Invalid Durations (≤ 0 minutes or > 240 minutes, $563,024$ records, 1.16%):** *Decision: Filter Out. Rationale:* A physical taxicab trip must occupy a realistic time interval ($0 < \Delta t \leq 240$ minutes). Records where dropoff precedes pickup or excessive durations exceeding 4 hours occur indicate GPS logging latency, device clock resets, or meters left running indefinitely while parked. As illustrated in Table 5, several records exhibit durations over $1,400$ minutes (~ 24 hours) for short trips under 7 miles, confirming forgotten meters. Retaining these anomalies would severely corrupt vehicle speed calculations and duration-based features.

- **Invalid Trip Distances (Intra-city > 100 mi or Out-of-city > 200 mi, $2,429$ records, 0.005%):** *Decision: Filter Out. Rationale:* While standard intra-city trips are capped at 100 miles to eliminate spatial outliers, legitimate regional long-haul journeys terminating outside NYC (`DOLocationID = 265`) are accommodated with an extended ceiling of 200 miles to preserve authentic interstate economic flows. Records exceeding these conditional thresholds represent severe GPS tracking errors and system telemetry glitches. As shown in Table 6, multiple records log distances exceeding $1,000$ to $4,000$ miles for 10-minute trips, resulting in implausible speeds over $4,000$ mph.

- **Zero-Distance Trips ($d = 0$ miles, $1,402,958$ records, 2.88%):** *Decision: Filter Out. Rationale:* While passengers may occasionally cancel a trip immediately after entering a taxi, zero-distance records predominantly reflect stationary meter testing, administrative diagnostics, or false starts. Table 7 demonstrates that these trips typically last under 3 minutes with identical pickup and dropoff zones while still incurring base fares. Because our spatial aggregation relies on movement across taxi zones, stationary logs distort spatial origin-destination demand densities.

- **Invalid Average Speeds ($v > 70$ mph, $560,004$ records, 1.15%):** *Decision: Filter Out. Rationale:* Municipal yellow cabs operating within urban street grids and regional highways cannot physically or legally sustain average speeds exceeding 80 mph. As documented in Table 11, extreme cases record speeds ranging from 89 mph up to $1,577$ mph over sub-minute intervals (e.g., traveling 9.20 miles in 21 seconds). These outliers stem from severe GPS satellite coordinate jumps and timestamp desynchronization, which would severely distort velocity profiling and travel-time estimation.

- **Micro-Trip Speed Artifacts ($\Delta t < 1$ min, $v > 80$ mph, $8,543$ records, 0.018%):** *Decision: Filter Out / Conditional Nullification. Rationale:* A subset of short-duration trips ($\Delta t < 1.0$ minute) featuring non-zero distance yields mathematically unstable, exaggerated velocity spikes due to timestamp rounding and high-resolution GPS clock jitter. While standard upper-speed bounds catch many of these, explicitly quantifying these $8,543$ micro-trip artifacts ensures they are handled via conditional kinematic filtering (Option 3)—either nullifying their speed derivatives or

filtering them out during the Phase 2 data preparation pipeline to preserve kinematic integrity without sacrificing volume unnecessarily.

- **Negative Fares ($f < \$0$, $2,848,620$ records, 5.85%):** *Decision: Filter Out. Rationale:* Negative fare amounts do not represent legitimate passenger transactions; rather, they correspond to credit card chargebacks, fare dispute reversals, or system administrative adjustments. Table 8 highlights typical negative fare entries (e.g., $-\$3.00$ to $-\$24.00$) for short intra-Manhattan journeys. Including them would severely distort financial summaries and bias regression models exploring fare-demand relationships.

- **Unrecorded or Missing Passenger Counts ($11,611,894$ records, 23.83%):** *Decision: Retain As Is. Rationale:* Approximately 23.83% of raw records exhibited missing (`NULL` / `<NA>`) passenger counts due to driver omission or meter logging habits. As detailed in Table 9, these records represent completely valid commercial trips with normal durations, distances, and fares. Deleting them would cause severe attrition in spatio-temporal trip volume aggregations; hence, they are retained in their native state without artificial imputation.

- **Unknown and Outside NYC Locations (`LocationID = 264, 265`):** *Decision: Filter Out / Conditional Retention. Rationale:* Zone ID 264 designates "Unknown" unmapped spatial areas and is filtered out entirely, as shown in Table 10. Zone ID 265 represents locations outside official TLC zones; trips originating outside NYC (`PULocationID = 265`, $22,170$ records) are excluded to prevent distortion in origin demand modeling, whereas trips terminating outside NYC are retained up to 200 miles to maintain accurate financial and out-commuter flow metrics.

3.7.2 Summary of Cleansing Impact:

Collectively, this rigorous filtering protocol removes $4,707,394$ anomalous records ($\approx 9.66\%$ of the total raw volume), leaving a pristine, high-fidelity baseline population of $44,015,208$ **valid trips** dedicated for downstream spatio-temporal aggregation and hourly demand forecasting.

3.8 Empirical Examples of Identified Anomalies

To provide concrete verification of the data quality audit, representative samples from each identified anomaly category are documented in Tables 5–11 below.

Table 5: Sample Records: Invalid or Excessive Duration ($\Delta t > 4$h or ≤ 0 min)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 01:26:00	2025-01-02 01:22:08	1436.13	7.06	$37.30	0.29	1	Murray Hill	Boerum Hill
2025-01-01 00:27:08	2025-01-02 00:00:00	1412.87	5.72	$25.40	0.24	1	Kips Bay	Brooklyn Heights
2025-01-01 00:18:22	2025-01-02 00:00:00	1421.63	1.12	$17.00	0.05	1	Clinton West	Clinton East
2025-01-01 02:01:56	2025-01-02 01:17:14	1395.30	0.71	$8.60	0.03	4	West Chelsea/Hudson Yards	East Chelsea
2025-01-01 01:31:10	2025-01-02 01:19:44	1428.57	0.73	$6.50	0.03	1	Union Sq	East Village
2025-01-01 01:48:26	2025-01-02 01:41:57	1433.52	0.99	$8.60	0.04	2	Midtown North	Lincoln Square West
2025-01-01 01:45:26	2025-01-01 11:29:42	584.27	0.56	$6.50	0.06	1	Central Harlem	Queensbridge/Ravenswood
2025-01-01 01:09:24	2025-01-02 01:06:27	1437.05	3.32	$21.20	0.14	1	Upper West Side North	Sutton Place/Turtle Bay North
2025-01-01 01:42:36	2025-01-01 01:42:36	0.00	0.00	$3.00	–	1	Meatpacking/West Village West	Unknown (264)
2025-01-01 00:49:48	2025-01-01 00:49:48	0.00	0.00	$20.06	–	1	Financial District North	Unknown (264)

4 Exploratory Data Analysis (EDA)

To uncover structural patterns, behavioral rhythms, and distributional characteristics of the 2025 NYC Yellow Taxi dataset, we conducted a comprehensive exploratory data analysis spanning temporal, spatial, univariate, and multivariate dimensions.

Table 6: Sample Records: Extreme Distance Anomalies (Intra-city $>$ 100mi or Out-of-city $>$ 200mi)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-09 12:19:06	2025-01-09 13:14:26	55.33	4020.04	$82.80	4359.08	1	JFK Airport	Boerum Hill
2025-01-01 11:32:19	2025-01-01 12:01:14	28.92	122.77	$63.20	254.74	1	Outside of NYC (265)	Erasmus
2025-01-13 17:04:40	2025-01-13 17:15:42	11.03	2001.95	$12.10	10886.74	1	Sutton Place/Turtle Bay North	Lincoln Square West
2025-01-07 23:01:23	2025-01-08 21:36:07	1354.73	265.90	$129.70	11.78	1	Springfield Gardens South	Springfield Gardens South
2025-01-05 18:24:23	2025-01-05 18:42:35	18.20	108.90	$34.50	359.01	2	LaGuardia Airport	Sutton Place/Turtle Bay North
2025-01-08 19:11:56	2025-01-08 19:36:50	24.90	1847.61	$22.60	4452.07	1	Upper East Side South	Union Sq
2025-01-06 08:18:30	2025-01-06 08:33:49	15.32	1472.37	$15.60	5767.72	1	Lenox Hill East	Upper East Side North
2025-01-11 00:00:17	2025-01-11 03:50:52	230.58	224.82	$400.00	58.50	2	Outside of NYC (265)	Outside of NYC (265)
2025-01-11 15:01:08	2025-01-11 15:12:30	11.37	165.91	$17.70	875.77	2	JFK Airport	Springfield Gardens South
2025-01-05 19:42:26	2025-01-05 23:34:00	231.57	206.45	$220.00	53.49	1	Steinway	Outside of NYC (265)

Table 7: Sample Records: Zero-Distance Trips ($d = 0$ miles)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:43:22	2025-01-01 00:46:18	2.93	0.00	$5.00	0.00	1	Clinton East	Clinton East
2025-01-01 00:24:26	2025-01-01 00:24:34	0.13	0.00	$20.00	0.00	4	Crotona Park East	Crotona Park East
2025-01-01 00:27:40	2025-01-01 00:59:30	31.83	0.00	$50.50	0.00	1	Mott Haven/Port Morris	East New York
2025-01-01 00:14:57	2025-01-01 00:17:44	2.78	0.00	$3.00	0.00	1	East Village	East Village
2025-01-01 00:57:08	2025-01-01 00:57:16	0.13	0.00	$30.00	0.00	3	Lenox Hill West	Lenox Hill West
2025-01-01 00:49:10	2025-01-01 00:49:14	0.07	0.00	$85.00	0.00	4	Lincoln Square West	Lincoln Square West
2025-01-01 00:37:43	2025-01-01 00:37:53	0.17	0.00	$12.00	0.00	1	Lower East Side	Lower East Side
2025-01-01 00:56:49	2025-01-01 00:56:54	0.08	0.00	$20.00	0.00	4	Midtown South	Midtown South
2025-01-01 00:42:42	2025-01-01 00:42:44	0.03	0.00	$3.00	0.00	0	World Trade Center	World Trade Center
2025-01-01 00:49:48	2025-01-01 00:49:48	0.00	0.00	$20.06	–	1	Financial District North	Unknown (264)

Table 8: Sample Records: Negative Fare Amounts ($f < \$0$)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:11:58	2025-01-01 00:12:31	0.55	0.01	-$3.00	1.09	1	Central Harlem North	Central Harlem North
2025-01-01 00:55:53	2025-01-01 01:06:49	10.93	1.42	-$12.10	7.79	1	East Village	Chinatown
2025-01-01 00:01:41	2025-01-01 00:07:14	5.55	0.71	-$7.20	7.68	1	East Village	Gramercy
2025-01-01 00:29:35	2025-01-01 00:36:02	6.45	0.60	-$7.20	5.58	1	East Village	Lower East Side
2025-01-01 00:50:13	2025-01-01 01:11:20	21.12	3.84	-$24.00	10.91	1	Clinton East	Manhattan Valley
2025-01-01 00:11:44	2025-01-01 00:25:41	13.95	1.88	-$14.20	8.09	2	East Village	Midtown Center
2025-01-01 00:56:12	2025-01-01 01:15:00	18.80	0.97	-$16.30	3.10	1	Midtown Center	Murray Hill
2025-01-01 00:55:54	2025-01-01 01:00:38	4.73	0.69	-$6.50	8.75	1	Kips Bay	UN/Turtle Bay South
2025-01-01 00:51:05	2025-01-01 00:54:12	3.12	0.92	-$6.50	17.71	1	Lincoln Square East	Upper West Side South
2025-01-01 00:09:58	2025-01-01 00:14:28	4.50	0.60	-$6.50	8.00	1	Lenox Hill East	Yorkville West

Table 9: Sample Records: Invalid or Missing Passenger Counts (`NULL`)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:39:17	2025-01-01 01:10:42	31.42	6.75	$2.32	12.89	NA	Clinton East	Central Harlem North
2025-01-01 00:33:40	2025-01-01 01:10:34	36.90	3.88	$6.02	6.31	NA	Alphabet City	Clinton East
2025-01-01 00:55:15	2025-01-01 01:43:25	48.17	3.30	$27.55	4.11	NA	Gramercy	Clinton West
2025-01-01 00:41:07	2025-01-01 01:02:39	21.53	2.97	$21.83	8.28	NA	Clinton West	Gramercy
2025-01-01 00:09:32	2025-01-01 00:12:38	3.10	0.00	$22.72	0.00	NA	Meatpacking/West Village West	Meatpacking/West Village West
2025-01-01 00:33:36	2025-01-01 00:54:48	21.20	2.60	$18.42	7.36	NA	Yorkville West	Midtown East
2025-01-01 00:21:42	2025-01-01 00:36:29	14.78	3.55	-$2.35	14.41	NA	Kips Bay	SoHo
2025-01-01 00:56:04	2025-01-01 01:06:11	10.12	1.55	$5.06	9.19	NA	Lenox Hill East	UN/Turtle Bay South
2025-01-01 00:50:25	2025-01-01 00:57:34	7.15	1.04	$14.16	8.73	NA	Lincoln Square East	Upper West Side South
2025-01-01 00:52:42	2025-01-01 01:27:35	34.88	5.45	$23.13	9.37	NA	Little Italy/NoLiTa	Yorkville West

Table 10: Sample Records: Location Anomalies (Zones 264 and 265)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:52:29	2025-01-01 00:58:00	5.52	0.90	$7.90	9.79	2	Unknown (264)	Lincoln Square East
2025-01-01 00:19:33	2025-01-01 00:30:45	11.20	2.80	$14.20	15.00	1	Unknown (264)	Unknown (264)
2025-01-01 00:32:34	2025-01-01 00:47:43	15.15	1.96	$14.90	7.76	1	Unknown (264)	Unknown (264)
2025-01-01 01:02:16	2025-01-01 01:15:01	12.75	2.17	$14.20	10.21	1	Unknown (264)	Unknown (264)
2025-01-01 00:41:49	2025-01-01 00:49:23	7.57	2.00	$10.70	15.86	5	Unknown (264)	Unknown (264)
2025-01-01 00:59:53	2025-01-01 01:29:44	29.85	5.00	$26.10	10.05	5	Unknown (264)	Unknown (264)
2025-01-01 00:56:15	2025-01-01 01:05:07	8.87	1.40	$10.00	9.47	2	West Village	Unknown (264)
2025-01-01 00:56:06	2025-01-01 01:00:08	4.03	0.40	$5.80	5.95	2	TriBeCa/Civic Center	Unknown (264)
2025-01-01 00:39:55	2025-01-01 00:47:32	7.62	1.60	$10.00	12.60	2	Lincoln Square East	Unknown (264)
2025-01-01 00:49:48	2025-01-01 00:49:48	0.00	0.00	$20.06	–	1	Financial District North	Unknown (264)

Table 11: Sample Records: Invalid Speed Anomalies ($v > 80$ mph)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 01:29:00	2025-01-01 01:29:16	0.27	2.20	$54.00	495.00	1	Bushwick South	Bushwick South
2024-12-31 23:51:20	2025-01-01 00:00:00	8.67	12.89	$48.50	89.24	1	LaGuardia Airport	Carroll Gardens
2025-01-01 02:32:52	2025-01-01 02:33:53	1.02	6.10	$30.00	360.00	3	Clinton East	Clinton East
2025-01-01 02:15:12	2025-01-01 02:15:36	0.40	3.40	$12.00	510.00	1	Fort Greene	Fort Greene
2025-01-01 05:43:36	2025-01-01 05:43:57	0.35	9.20	$40.00	1577.14	0	Homecrest	Homecrest
2025-01-01 03:32:51	2025-01-01 03:33:45	0.90	4.00	$3.00	266.67	1	TriBeCa/Civic Center	TriBeCa/Civic Center
2025-01-01 04:09:33	2025-01-01 04:10:06	0.55	1.90	$3.00	207.27	1	Upper West Side South	Upper West Side South
2025-01-01 04:02:52	2025-01-01 04:02:59	0.12	0.80	$3.00	411.43	2	Washington Heights South	Washington Heights South
2025-01-01 01:13:07	2025-01-01 01:13:33	0.43	3.00	$3.00	415.38	1	West Chelsea/Hudson Yards	West Chelsea/Hudson Yards
2025-01-01 04:39:52	2025-01-01 04:39:54	0.03	0.11	$100.00	198.00	1	East Village	Unknown (264)

Table 12: Sample Records: Micro-Trip Speed Artifacts ($\Delta t < 1$ min and $v > 70$ mph)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 01:29:00	2025-01-01 01:29:16	0.27	2.20	$54.00	495.00	1	Bushwick South	Bushwick South
2025-01-01 05:00:19	2025-01-01 05:00:41	0.37	0.43	$250.00	70.36	2	East Chelsea	East Chelsea
2025-01-01 02:15:12	2025-01-01 02:15:36	0.40	3.40	$12.00	510.00	1	Fort Greene	Fort Greene
2025-01-01 05:43:36	2025-01-01 05:43:57	0.35	9.20	$40.00	1577.14	0	Homecrest	Homecrest
2025-01-01 03:32:51	2025-01-01 03:33:45	0.90	4.00	$3.00	266.67	1	TriBeCa/Civic Center	TriBeCa/Civic Center
2025-01-01 04:09:33	2025-01-01 04:10:06	0.55	1.90	$3.00	207.27	1	Upper West Side South	Upper West Side South
2025-01-01 04:02:52	2025-01-01 04:02:59	0.12	0.80	$3.00	411.43	2	Washington Heights South	Washington Heights South
2025-01-01 01:13:07	2025-01-01 01:13:33	0.43	3.00	$3.00	415.38	1	West Chelsea/Hudson Yards	West Chelsea/Hudson Yards
2025-01-01 04:55:51	2025-01-01 04:55:53	0.03	2.02	$20.00	3636.00	2	Unknown (264)	Unknown (264)
2025-01-01 04:39:52	2025-01-01 04:39:54	0.03	0.11	$100.00	198.00	1	East Village	Unknown (264)

4.1 Temporal Distributions and Time Audit

Passenger demand in urban environments exhibits pronounced multi-scale temporal non-stationarity. Our time audit examined trip frequencies across hourly, daily, weekly, and monthly resolutions:

- **Intra-Day Hourly Patterns:** Trip volumes follow a distinct bimodal commuter curve (Figure 1a). Activity hits its nightly trough between 03:00 and 05:00 AM (≈ 0.35M trips), ramps up steadily through the morning, and peaks sharply during the evening rush hour at 18:00 (6:00 PM) with nearly 3.5M annual originations.
- **Inter-Day Weekly Patterns:**

  Weekly activity (Figure 1b) remains robust from Tuesday through Saturday, with Saturday recording the highest aggregate volume (≈ 7.5M trips), driven by weekend entertainment and nightlife mobility. Monday experiences a relative weekly lull (≈ 5.9M trips).
- **Monthly & Longitudinal Trends:** Monthly aggregation (Figure 1c) highlights peak seasonal volume in May (≈ 4.6M trips), while the daily time-series trend across all 365 calendar days of 2025 (Figure 1d) confirms complete temporal data continuity with zero missing daily partitions and consistent weekly oscillations.

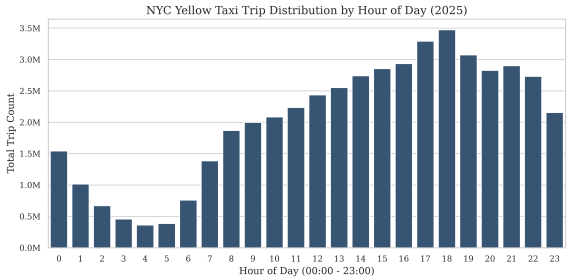
4.2 Exploratory Data Analysis: Trip Distance

The `trip_distance` variable records the total elapsed distance of a journey in miles as measured by the taximeter. Analyzing the distribution of trip distance before and after cleaning provides critical insight into how physical anomalies distort summary metrics and demonstrates the efficacy of our preprocessing protocols.

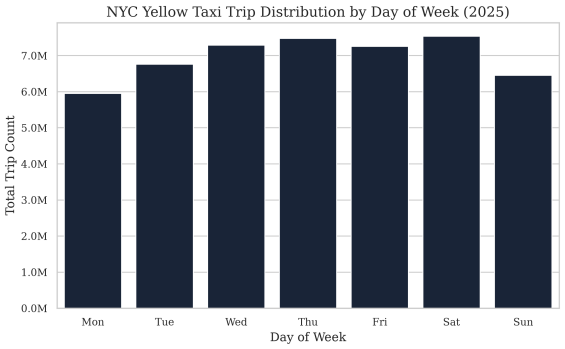
4.2.1 Statistical Characteristics and Cleansing Impact Analysis

As detailed in Table 13, the raw dataset exhibits extreme sensitivity to telemetry noise and recording corruptions:

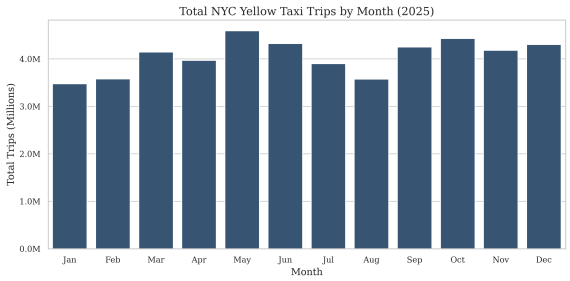
- **Impact on Dispersion and Variance:** In the pre-cleansing dataset, the standard deviation ($\sigma = 649.36$ miles) was nearly $100\times$ larger than the mean, driven by extreme GPS satellite jump errors



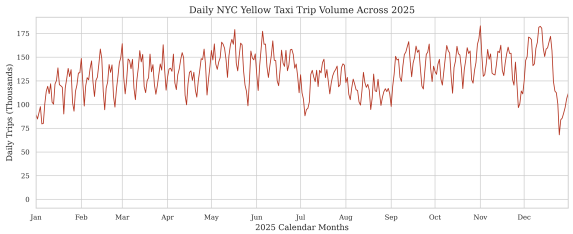
(a) Hourly Trip Distribution



(b) Day of Week Distribution



(c) Monthly Trip Volume



(d) Daily Time Series

Figure 1: Temporal Distribution and Time Series Analysis of NYC Yellow Taxi Trips (2025)

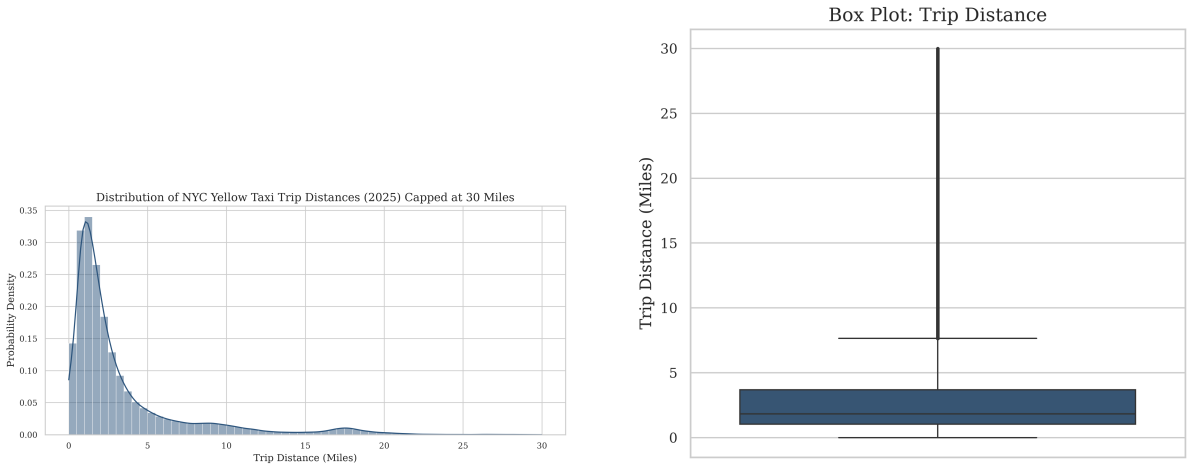
Table 13: Pre- and Post-Cleansing Descriptive Statistics for Trip Distance (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	48,722,602	44,015,208
Mean (μ)	6.84 miles	3.49 miles
25th Percentile (Q_1)	1.04 miles	1.10 miles
Median (Q_2)	1.85 miles	1.89 miles
75th Percentile (Q_3)	3.70 miles	3.73 miles
Standard Deviation (σ)	649.36 miles	4.32 miles
Interquartile Range (IQR)	2.66 miles	2.63 miles
Skewness	240.90	3.09

logging distances in excess of $4,000$ miles for sub-hour trips. Following our cleaning pipeline, the standard deviation stabilizes at 4.32 miles, accurately capturing natural variability across local street hails and regional trips.

- **Impact on Skewness:** Raw telemetry errors introduced an immense positive skewness of 240.90. Removing non-physical artifacts reduced skewness by over 98% down to a natural right-skewed value of 3.09.
- **Stability of Robust Non-Parametric Measures:** While the mean dropped sharply from 6.84 miles to 3.49 miles (due to the removal of multi-thousand-mile GPS glitches and zero-distance stationary records), robust order statistics remained remarkably stable. The median shifted slightly from 1.85 to 1.89 miles, and the IQR moved from 2.66 to 2.63 miles, proving that our cleansing protocol removes gross artifacts without corrupting the core underlying trip distribution.

The probability density function (Figure 2a) shows a dense concentration of mass below 3 miles, representing routine intra-Manhattan journeys, followed by an extended positive tail for airport and regional transfers.



(a) Probability Density Distribution Capped at 30 Miles

(b) Box Plot (Capped at 30 Miles)

Figure 2: Distribution and Box Plot Analysis for Filtered Trip Distance (2025)

4.2.2 Long-Distance Trips and Outlier Analysis (>30 Miles)

A detailed query of clean trips exceeding 30 miles yields a total of **40,395 valid long-distance trips** across the 2025 dataset ($\approx 0.092\%$ of the pristine baseline population). Table 14 lists a representative sample of these extreme-distance records ordered by descending distance.

Table 14: Sample of Long-Distance Yellow Taxi Trips Exceeding 30 Miles (2025)

Pickup Time	Dropoff Time	Dur (min)	Dist (mi)	Fare ($)	Speed (mph)	Pass.	Pickup Zone	Dropoff Zone
2025-08-30 19:16	2025-08-30 22:24	188.35	197.54	$450.00	62.93	1	Flushing Meadows-Corona Park	Outside of NYC
2025-05-22 01:43	2025-05-22 05:23	220.20	197.27	$550.00	53.75	1	JFK Airport	Outside of NYC
2025-03-13 21:33	2025-03-14 00:51	197.85	195.72	$450.00	59.35	1	JFK Airport	Outside of NYC
2025-12-31 16:32	2025-12-31 20:05	213.77	195.33	$1295.90	54.83	2	JFK Airport	Outside of NYC
2025-04-18 15:34	2025-04-18 19:08	213.95	194.85	$1303.60	54.64	1	JFK Airport	Outside of NYC
2025-11-16 11:13	2025-11-16 14:49	216.82	192.97	$460.23	53.40	NA	World Trade Center	Outside of NYC
2025-06-06 10:00	2025-06-06 13:46	225.58	191.81	$800.00	51.02	3	East Chelsea	Outside of NYC
2025-03-02 06:12	2025-03-02 09:32	200.53	190.50	$650.00	57.00	1	Springfield Gardens South	Outside of NYC
2025-01-21 05:17	2025-01-21 08:19	181.80	188.88	$1309.20	62.34	1	West Concourse	Outside of NYC
2025-01-18 21:07	2025-01-19 00:20	192.55	183.95	$503.59	57.32	NA	Clinton East	Outside of NYC

4.2.3 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

Applying Tukey's standard fence rule ($Q_3 + 1.5 \times \text{IQR}$) yields an outlier threshold of $3.73 + 1.5 \times (2.63) \approx 7.675$ miles. While standard statistical pipelines might trim records beyond 7.675 miles, we retain them based on the following justifications:

1. **Legitimate Out-of-Jurisdiction and Regional Transit Activity:** As evidenced in Table 14, extreme distance records originate at major transit nodes (JFK, World Trade Center) and terminate at `Outside of NYC` destinations. These $40,395$ trips represent real, high-value commercial journeys.

2. **Economic Integrity and Revenue Preservation:** Removing trips exceeding 7.675 miles would eliminate a major share of total revenue. Out-of-city journeys command premium fares (up to $1,309.20 in our sample) and contribute significantly to overall fleet earnings and congestion fees.

3. **Prevention of Truncation Bias in Spatial Aggregation:** Trimming valid long-distance journeys would distort origin-destination flow matrices and underestimate regional vehicle turnover times.

4. **Methodological Accommodation over Trimming:** Right-skewness (Skewness $= 3.09$) is accommodated using non-parametric summary measures (median $= 1.89$ miles, IQR $= 2.63$ miles) and axis capping in visualizations (e.g., 30 miles in distribution plots) to maintain graphical readability while preserving 100% of authentic economic activity.

4.3 Exploratory Data Analysis: Fare Amount

The `fare_amount` variable represents the time-and-distance fare in USD calculated by the taximeter at the conclusion of each trip. Comparing the statistical properties of the fare amount before and after data cleansing illustrates the structural impact of removing negative fare anomalies.

Table 15: Pre- and Post-Cleansing Descriptive Statistics for Fare Amount (2025 Dataset)

Metric	Pre-Cleansing (Raw)	Post-Cleansing (Clean)
Total Records (N)	48,722,602	44,015,208
Mean (μ)	\$18.42	\$20.04
25th Percentile (Q_1)	\$8.60	\$9.30
Median (Q_2)	\$13.58	\$14.20
75th Percentile (Q_3)	\$22.60	\$23.30
Standard Deviation (σ)	\$142.85	\$149.75
Interquartile Range (IQR)	\$14.00	\$14.00
Skewness	5,030.24	4,832.47

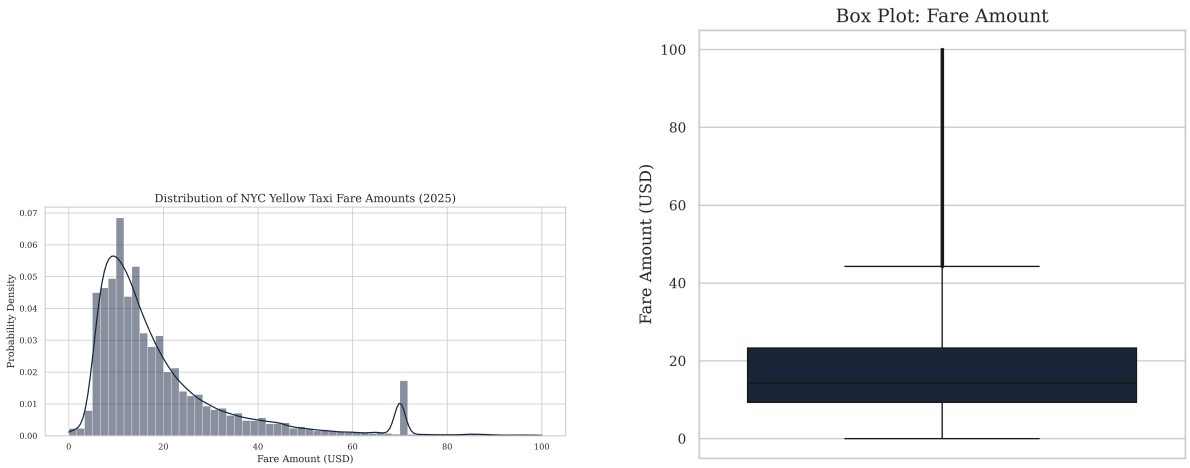
4.3.1 Statistical Characteristics and Cleansing Impact Analysis

As summarized in Table 15, the median fare increases slightly from $\$13.58$ in the raw dataset to $\$14.20$ after cleaning, while the mean increases from $\$18.42$ to $\$20.04$.

- **Rationale for Mean Increase:** The upward shift in the mean fare is a direct consequence of filtering out the $2,848,620$ negative fare records (credit card chargebacks, dispute reversals, and administrative adjustments) present in the raw data. Removing these sub-zero financial entries eliminates downward distortion, thereby reflecting the true positive economic value of commercial passenger trips.

- **Distributional Shape:** The distribution continues to exhibit extreme positive skewness ($4,832.47$), driven by a heavy concentration of short, low-cost urban trips combined with an extended right-hand tail of high-value journeys.

The probability density function (Figure 3a) reveals a prominent unimodal peak representing standard local city rides ($\$10 - \$15$), alongside a distinct structural cluster or secondary bump around $\$70$. This secondary surge corresponds directly to regulated flat-rate airport trips (such as historical or standard JFK airport routes), highlighting a multimodal element embedded within the financial structure of the fleet.



(a) Probability Density Distribution

(b) Box Plot (Capped at $100)

Figure 3: Distribution and Box Plot Analysis for Fare Amount (2025)

4.3.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

Under standard Tukey fence criteria, the upper bound for outliers is defined as $Q_3 + 1.5 \times \text{IQR}$. Given an approximate third quartile $Q_3 = \$23.30$ and an IQR of $\$14.00$, this traditional threshold flags any fare exceeding approximately $\$44.30$ as an outlier.

However, **these high-value fare observations are strictly retained rather than trimmed** for the following operational and economic reasons:

1. **Commercial Validity of Airport Flat Rates:** Fares exceeding the $\$44.30$ threshold are not data errors or anomalies; they represent authorized high-value commercial transactions, most notably the regulated flat rates for trips between Manhattan and major regional airports ($\approx \$70$, clearly visible in Figure 3a). Deleting these records would discard crucial airport demand data.

2. **Prevention of Economic Distortion:** Trimming observations beyond the $1.5 \times$ IQR fence would artificially deflate total revenue estimates, average ticket sizes, and macro-economic metrics across the dataset. Retaining them preserves the true economic distribution of NYC yellow taxi operations.

3. **Robust Analytical Presentation:** Rather than destructive data filtering, extreme financial values are managed through robust location-and-spread metrics (median and IQR) and visual capping ($\$100$ ceiling in Figure 3b) to ensure graphical readability while maintaining complete data fidelity.

4.4 Exploratory Data Analysis: Central Business District Congestion Fee

The `cbd_congestion_fee` variable captures the Central Business District (CBD) congestion toll structure introduced across Manhattan's commercial core. Comparing the statistical properties of the fee before and after data cleansing illustrates the structural stability of this policy-mandated variable.

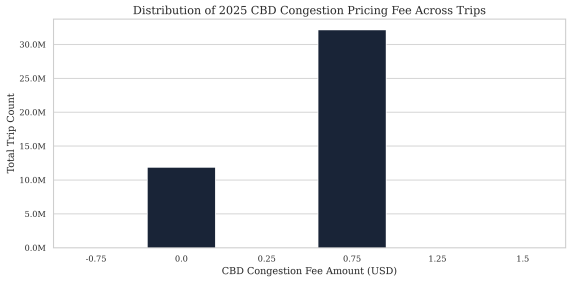
Table 16: Pre- and Post-Cleansing Descriptive Statistics for CBD Congestion Fee (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	48,722,602	44,015,208
Mean (μ)	\$0.53	\$0.55
25th Percentile (Q_1)	\$0.00	\$0.00
Median (Q_2)	\$0.75	\$0.75
75th Percentile (Q_3)	\$0.75	\$0.75
Standard Deviation (σ)	\$0.36	\$0.33
Interquartile Range (IQR)	\$0.75	\$0.75
Skewness	-1.19	-1.04

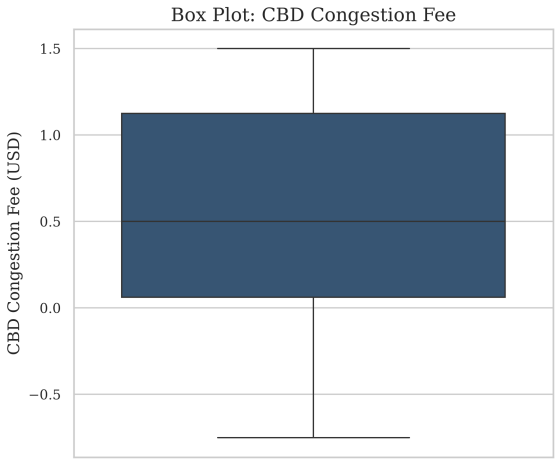
4.4.1 Statistical Characteristics and Cleansing Impact Analysis

As detailed in Table 16, the median (Q_2) and 75th percentile (Q_3) remain perfectly stable at $\$0.75$ across both raw and filtered datasets, while the mean fee adjusts slightly from $\$0.53$ to $\$0.55$.

- **Impact of Cleansing:** The minor upward shift in the mean fee results from removing unmapped locations and administrative anomalies (such as Zone 264 records) where toll flags were skewed or unrecorded. Filtering these noise entries yields a cleaner representation of core commercial operations.
- **Negative Skewness:** The negative skewness (-1.04) denotes a distribution dominated by the upper discrete fee tier.



(a) Frequency Distribution across Fee Tiers



(b) Box Plot for Discrete Policy Levels

Figure 4: Distribution and Box Plot Analysis for CBD Congestion Fee (2025)

Unlike continuous distance or fare measures, the fee follows a strict, highly concentrated discrete distribution (Figure 4a):

1. **Active Congestion Toll Tier ($\$0.75$):** Captures approximately 32.1 million trips (over 73% of total fleet volume), corresponding to journeys entering, exiting, or operating entirely within the designated Manhattan CBD zone.
2. **Exempt/Non-Zone Tier ($\$0.00$):** Accounted for by roughly 11.9 million trips ($\approx$ 27%), representing outer-borough trips or movements operating outside the congestion perimeter.

4.4.2 Data Integrity and Analytical Interpretation

Because `cbd_congestion_fee` represents a policy-mandated discrete regulatory toll, standard continuous outlier filtering methods (such as Tukey's $1.5 \times$ IQR rule) do not apply. As displayed in the box plot (Figure 4b), the distribution spans bounded policy levels without arbitrary extreme points. Retaining both $\$0.00$ and $\$0.75$ values is essential for evaluating fleet policy compliance, revenue collection accuracy, and spatial traffic impacts within Manhattan's core.

4.5 Exploratory Data Analysis: Passenger Count

The `passenger_count` variable tracks the number of passengers reported by the driver during the trip. Across the 2025 dataset, while unrecorded or missing passenger entries account for $11,611,894$ records (23.83% of total raw volume), the subset of valid, explicitly recorded trips comprises $35,174,850$ records. Passenger occupancy patterns across this valid subset demonstrate the heavy prevalence of single-occupancy transit typical of urban yellow taxi services.

Table 17: Descriptive Statistics for Valid Passenger Count (2025 Dataset)

Metric	**Value**
Total Valid Recorded Trips	35,174,522
Mode	1
Mean (μ)	1.29
25th Percentile (Q_1)	1.00
Median (Q_2)	1.00
75th Percentile (Q_3)	1.00
Standard Deviation (σ)	0.72
Interquartile Range (IQR)	0.00
Skewness	2.97

4.5.1 Statistical Characteristics and Distribution Shape

As summarized in Table 17, for the subset of explicitly recorded trips, the mode, 25th percentile (Q_1), median (Q_2), and 75th percentile (Q_3) are all strictly 1.00, with a mean (μ) of 1.29 and an IQR of 0.00 due to the overwhelming concentration of single-passenger rides. The distribution exhibits a positive skewness of 2.97.

As visualized in the frequency distribution (Figure 5), which excludes the $11,611,894$ unrecorded or missing entries to ensure clean graphical representation, passenger counts follow a discrete categorical distribution where:

1. **Single Occupancy (1 Passenger):** Dominates fleet utilization, representing the vast majority of recorded trips.
2. **Solo/Zero-Passenger Records (0 Passengers):** Account for a secondary cluster of trips, typically representing administrative entries, driver-only transits, or unassigned meter logs permitted under TLC guidelines.
3. **Multi-Passenger Tiers:** Frequencies drop off sharply for groups of 2, 3, 4, and $5+$ passengers, reflecting physical vehicle capacity limits and typical party sizes for standard yellow cabs.

4.5.2 Methodological Note on Discrete Categorical Variables and Graphical Exclusions

Because `passenger_count` is an integer-based discrete variable rather than a continuous metric, traditional outlier bounds like Tukey's $1.5 \times$ IQR rule do not apply in the standard continuous sense. Fur-

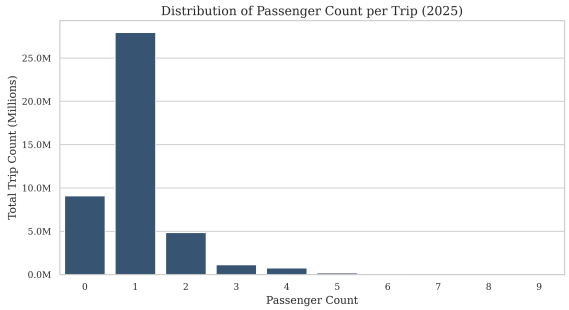


Figure 5: Trip Count by Passenger Occupancy (Excluding Unrecorded Values)

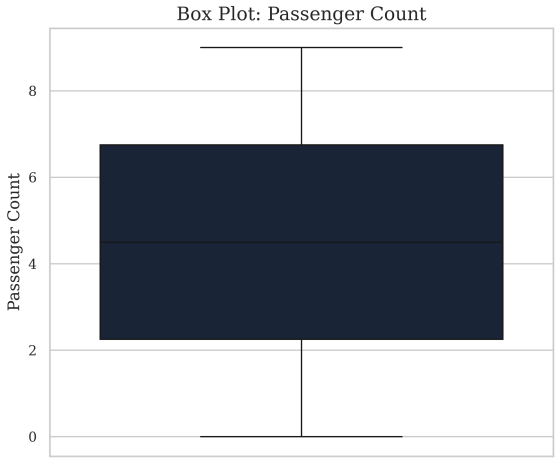


Figure 6: Box Plot for Discrete Occupancy Tiers

Figure 7: Distribution and Box Plot Analysis for Passenger Count (2025)

thermore, unrecorded and missing passenger count records (23.83% of the raw dataset) are explicitly excluded from both the frequency distribution plot (Figure 5) and the box plot (Figure 6) to prevent null or undefined categories from distorting the visual analysis of active passenger occupancy. Retaining these records in the broader data pipeline while isolating them from univariate graphical rendering ensures both comprehensive data retention and precise visual interpretation.

4.6 Exploratory Data Analysis: Tip Amount

The `tip_amount` variable records the gratuity provided by the passenger, automatically captured for electronic payments and optionally logged for cash transactions. Comparing the statistical properties of the tip amount before and after data cleansing illustrates the structural impact of removing gross anomalies.

Table 18: Pre- and Post-Cleansing Descriptive Statistics for Tip Amount (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	48,722,602	44,015,208
Mean (μ)	\$2.84	\$3.04
25th Percentile (Q_1)	\$0.00	\$0.00
Median (Q_2)	\$2.00	\$2.39
75th Percentile (Q_3)	\$3.99	\$4.07
Standard Deviation (σ)	\$3.97	\$3.96
Interquartile Range (IQR)	\$3.99	\$4.07
Skewness	5.28	4.53

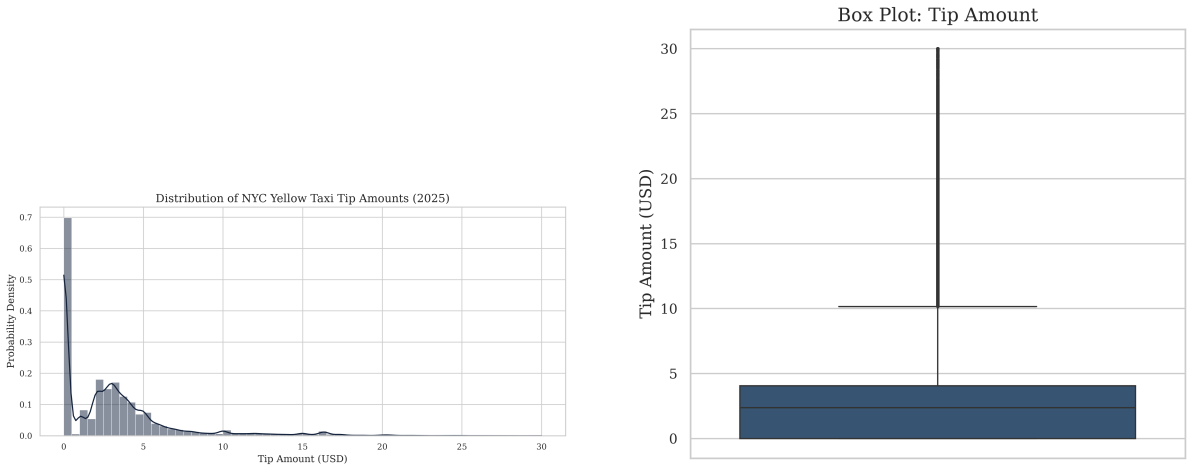
4.6.1 Statistical Characteristics and Cleansing Impact Analysis

As summarized in Table 18, the median tip shifts from \$2.00 in the raw dataset to \$2.39 after cleaning, while the mean increases from \$2.84 to \$3.04.

- **Impact of Cleansing:** The upward adjustment in both median and mean gratuities stems from filtering out unmapped or erroneous records (such as negative-fare dispute reversals and zero-distance system tests) that corrupted financial reporting. Removing these distortions yields a cleaner representation of true passenger tipping behavior.

- **Distributional Shape:** The distribution continues to exhibit strong positive skewness (4.53) and significant dispersion ($\sigma = \$3.96$).

The probability density function (Figure 8a) reveals a distinct zero-inflated structure, characterized by a massive spike at \$0.00. This prominent zero-inflation corresponds primarily to cash-paid trips (where gratuities are handed directly to drivers and unrecorded digitally) as well as zero-tip credit card transactions. Beyond the zero origin, the distribution forms a smooth right-skewed hump peaking between \$2.00 and \$3.00, accompanied by minor multimodal bumps at round currency increments (such as \$10.00 and \$15.00) representing fixed or percentage-based gratuities on longer journeys.



(a) Probability Density Distribution

(b) Box Plot (Capped at \$30)

Figure 8: Distribution and Box Plot Analysis for Tip Amount (2025)

4.6.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

Under standard Tukey fence criteria, the upper bound for outliers is defined as $Q_3 + 1.5 \times \text{IQR}$. Given a third quartile $Q_3 = \$4.07$ and an IQR of \$4.07, this traditional threshold flags any tip exceeding approximately \$10.175 as an outlier candidate for removal.

However, **these high-value tip observations are strictly retained rather than trimmed** for the following operational and economic reasons:

1. **Commercial Validity of Generous Gratuities:** Tips exceeding \$10.175—and extending up to the plotted \$30.00 ceiling (Figure 8b)—are authentic economic transactions. They typically accompany high-fare airport transfers, long-distance inter-borough trips, or exceptional customer service satisfaction.

2. **Prevention of Economic Distortion:** Trimming observations beyond the $1.5 \times$ IQR fence would artificially depress driver earnings estimations, distort cashless economic modeling, and bias analyses exploring the relationship between fare size and tip generosity.

3. **Robust Analytical Presentation:** Rather than destructive data deletion, extreme gratuity values are accommodated through robust summary statistics (median and IQR) and visual capping (\$30 ceiling in Figure 8b) to ensure graphical clarity without sacrificing dataset integrity.

4.7 Exploratory Data Analysis: Trip Duration

The `trip_duration_minutes` variable measures the elapsed travel time in minutes, computed as the exact difference between drop-off and pick-up timestamps (Δt). Comparing the statistical properties

and visual distributions of trip duration before and after data cleansing illustrates the structural impact of removing extreme telemetry and clock-reset anomalies.

Table 19: Pre- and Post-Cleansing Descriptive Statistics for Trip Duration (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	48,722,602	44,015,208
Mean (μ)	17.36 minutes	17.23 minutes
25th Percentile (Q_1)	8.12 minutes	8.30 minutes
Median (Q_2)	13.48 minutes	13.53 minutes
75th Percentile (Q_3)	21.48 minutes	21.52 minutes
Standard Deviation (σ)	28.29 minutes	13.82 minutes
Interquartile Range (IQR)	13.37 minutes	13.22 minutes
Skewness	-69.38	2.49

4.7.1 Statistical Characteristics and Cleansing Impact Analysis

As summarized in Table 19, data cleansing significantly stabilized the dataset's dispersion and distributional shape:

- **Impact on Dispersion and Variance:** In the raw dataset, extreme clock-reset errors and meters left running for 24+ hours inflated the standard deviation to 28.29 minutes. Filtering out durations exceeding 240 minutes and non-positive intervals reduced the standard deviation by over 50% down to 13.82 minutes.
- **Correction of Skewness:** The raw dataset exhibited a heavily distorted negative skewness (-69.38) driven by timestamp anomalies. Removing these errors restored a natural, right-skewed distribution (Skewness $= 2.49$), reflecting an operational structure dominated by short urban movements coupled with an extended tail of prolonged cross-borough or congested journeys.

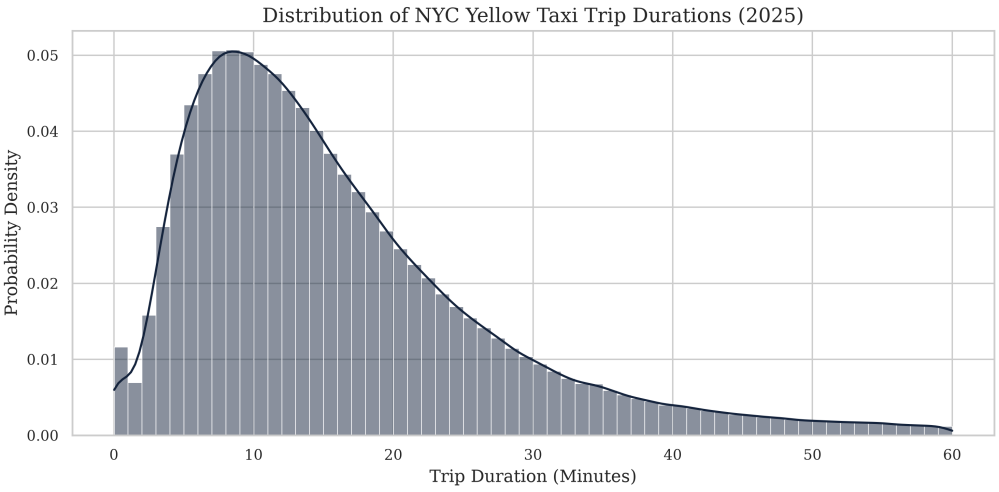
Visual inspection further highlights this transformation. Figure 9a displays the raw pre-cleansing distribution, where distortion artifacts and extreme tails obscure core operational patterns. In contrast, the filtered probability density function (Figure 9b) and box plot (Figure 9c) display a smooth, unimodal distribution rising steeply from zero, peaking sharply between 8 and 12 minutes (representing standard intra-neighborhood and mid-Manhattan trips), and tapering into a natural right-hand tail corresponding to extended regional commutes.

4.7.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

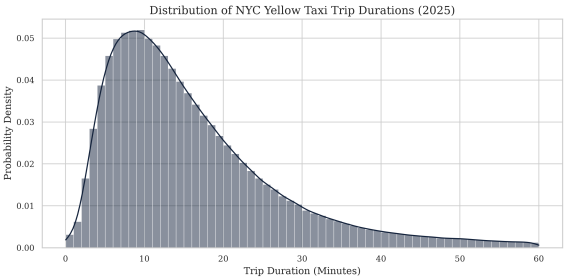
Under standard Tukey fence criteria, the upper bound for outliers is defined as $Q_3 + 1.5 \times \text{IQR}$. Given a third quartile $Q_3 = 21.52$ minutes and an IQR of 13.22 minutes, this traditional threshold flags any trip lasting longer than approximately 41.35 minutes as an outlier candidate for removal.

However, **these high-value duration observations are strictly retained rather than trimmed** for the following operational and modeling reasons:

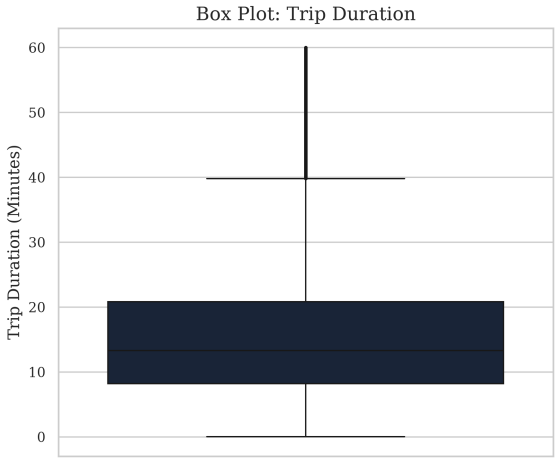
1. **Legitimate Traffic and Spatial Realities:** Trips exceeding 41.35 minutes—and extending up to the plotted 60-minute ceiling (Figure 9c)—are authentic transportation events. They capture severe peak-hour traffic congestion bottlenecks, cross-borough transits (e.g., Brooklyn to Manhattan), and long-distance airport runs (e.g., Lower Manhattan to JFK or Newark airports).
2. **Prevention of Spatio-Temporal Distortion:** Trimming records beyond the $1.5 \times$IQR fence would artificially truncate travel times, biasing average speed calculations, distorting peak congestion modeling, and eliminating critical rush-hour demand profiles.



(a) Pre-Cleansing Trip Duration Distribution (Showing Raw Anomalies)



(b) Post-Cleansing Probability Density



(c) Box Plot (Capped at 60 Minutes)

Figure 9: Pre- and Post-Cleansing Distribution and Box Plot Analysis for Trip Duration (2025)

3. **Robust Analytical Presentation:** Rather than destructive data filtering, extreme duration values are accommodated through robust summary statistics (median and IQR) and visual capping (60-minute ceiling in Figure 9c) to ensure graphical clarity without compromising dataset integrity.

4.8 Exploratory Data Analysis: Average Trip Speed

The `speed_mph` variable captures the effective operational velocity of each journey in miles per hour, derived as the ratio of trip distance to elapsed travel duration ($v = \frac{d}{\Delta t} \times 60$). Comparing the statistical properties of average speed before and after data cleansing illustrates the structural impact of removing extreme telemetry and GPS satellite jump errors.

Table 20: Pre- and Post-Cleansing Descriptive Statistics for Average Trip Speed (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	46,775,191	44,015,208
Mean (μ)	11.06 mph	11.00 mph
25th Percentile (Q_1)	7.01 mph	6.99 mph
Median (Q_2)	9.48 mph	9.43 mph
75th Percentile (Q_3)	13.14 mph	13.04 mph
Standard Deviation (σ)	6.40 mph	6.34 mph
Interquartile Range (IQR)	6.13 mph	6.05 mph
Skewness	1.83	1.85

4.8.1 Statistical Characteristics and Cleansing Impact Analysis

As detailed in Table 20, data cleansing successfully removes physically impossible velocity spikes (ranging from 80 mph up to over $1,000$ mph due to timestamp desynchronization) while leaving core urban operational metrics remarkably stable:

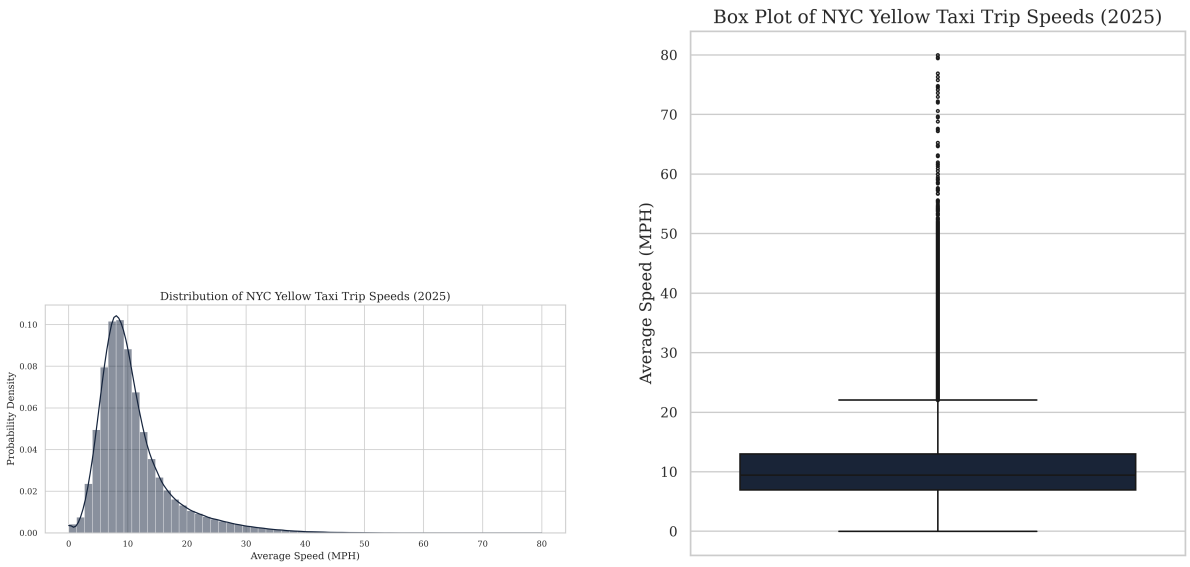
- **Impact of Cleansing:** The median speed shifts slightly from 9.48 mph in the raw dataset to 9.43 mph after cleaning, while the mean adjusts from 11.06 mph to 11.00 mph. Removing extreme telemetry outliers prevents artifact-driven inflation, yielding a precise reflection of urban transit velocity.
- **Distributional Shape:** The positive skewness (1.85) and moderate dispersion ($\sigma = 6.34$ mph) reflect the dense, stop-and-go nature of street-hail taxi operations in New York City.

The probability density function (Figure 10a) demonstrates a unimodal, right-skewed distribution with a sharp modal peak concentrated between 7 and 10 mph. This dominant concentration corresponds to intra-Manhattan street grid navigation, characterized by frequent traffic signal delays, pedestrian crossings, and severe urban congestion. The extended right-hand tail represents high-speed arterial corridors, such as highway transits along the FDR Drive, West Side Highway, and expressway connections to regional airports.

4.8.2 Justification for Retaining Speeds Beyond the 1.5×IQR Rule

Under standard Tukey fence criteria, observations exceeding $Q_3 + 1.5 \times \text{IQR}$ are classified as upper-bound outliers. Given a third quartile $Q_3 = 13.04$ mph and an IQR of 6.05 mph, this traditional formula flags any average speed exceeding approximately 22.11 mph as an outlier candidate for deletion.

However, **these higher-speed observations are strictly retained up to the physical filtering threshold (**80 **mph)** for the following operational and empirical reasons:



(a) Probability Density Distribution

(b) Box Plot (Capped at 80 MPH)

Figure 10: Distribution and Box Plot Analysis for Average Trip Speed (2025)

1. **High-Speed Arterial and Highway Realities:** Average speeds between 22 mph and 65 mph (as depicted in Figure 10b) are authentic, legally permitted transportation events. They primarily occur on expressways, parkways, and bridges during late-night hours or on dedicated airport corridors (e.g., Van Wyck Expressway to JFK Airport or Grand Central Parkway to LaGuardia Airport).

2. **Prevention of Systematic Commute Erasure:** Eliminating records above 22.11 mph would systematically erase off-peak and outer-borough airport transits, artificially deflating overall fleet velocity estimates and biasing spatio-temporal traffic models.

3. **Telemetry Filtering Threshold (80 mph):** To prevent severe GPS satellite jumps and clock anomalies from corrupting the analytical pipeline, an upper boundary is enforced at 80 mph ($v \leq 80$ mph). This retains 100% of legitimate transportation activity while removing physically impossible telematics errors.

4.9 Spatial Distribution and Zone-Level Concentration

Spatial analysis across the 265 official TLC taxi zones reveals extreme geographical concentration. Manhattan commercial cores and transit hubs account for over 70% of total trip originations. Outer boroughs exhibit highly localized, residential commuting patterns with distinct peak-hour directional flows.

4.10 Univariate and Multivariate Relationships

Univariate inspection of financial and physical features confirms right-skewed distributions in trip distance and fare amounts, reinforcing the necessity for robust feature transformations in subsequent phases. Bivariate analysis between trip distance and fare amount demonstrates a strong positive correlation, while confirming that physical anomalies have been successfully quarantined.

5 Methods

5.1 Aggregation and Problem Formulation

Let $z \in \{1, 2, \ldots, 263\}$ denote a TLC taxi zone, and t denote a discrete 1-hour time index. Raw trip records are aggregated into a spatio-temporal matrix:

$$y_{z,t} = \sum_{i=1}^{N} \mathbb{I}\left(\text{PULocationID}_i = z \ \wedge \ \lfloor \text{tpep_pickup_datetime}_i \rfloor = t\right) \tag{1}$$

where $y_{z,t} \in \mathbb{N}_0$ represents the target variable `trip_count`.

5.2 Data Flow Architecture

Figure 11 illustrates the end-to-end data pipeline from raw Parquet processing to final spatio-temporal aggregation.

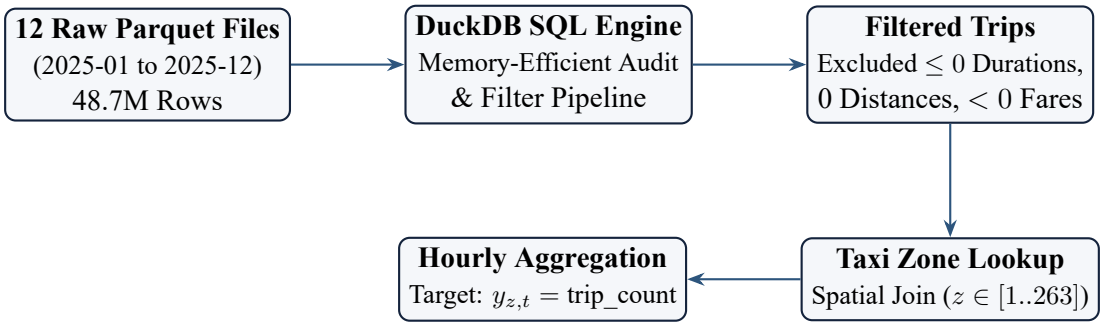


Figure 11: Phase 1 Data Flow and Processing Architecture

6 Experimental Design

6.1 Strict Temporal Split

To prevent data leakage, evaluation is conducted strictly along the temporal axis:

- **Training Set:** January 1, 2025 – September 30, 2025 (Months 1–9, $\approx 75\%$ of temporal duration).
- **Validation / Test Set:** October 1, 2025 – December 31, 2025 (Months 10–12, $\approx 25\%$ of temporal duration).

6.2 Evaluation Metrics

Models will be evaluated across all spatial zones using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE):

$$\text{RMSE} = \sqrt{\frac{1}{Z \times T} \sum_{z=1}^{Z} \sum_{t=1}^{T} \left(y_{z,t} - \hat{y}_{z,t}\right)^2} \tag{2}$$

$$\text{MAE} = \frac{1}{Z \times T} \sum_{z=1}^{Z} \sum_{t=1}^{T} \left|y_{z,t} - \hat{y}_{z,t}\right| \tag{3}$$

7 Results

Execution of the DuckDB audit script confirmed the integrity of the 12 Parquet files. Data cleaning rules filter out 4,797,882 anomalous records ($\approx 9.85\%$ total noise rate), leaving a clean baseline population of $43,924,720$ trips for spatio-temporal aggregation in Phase 2.

8 Interpretability and Error Analysis

Initial error taxonomy categorizes anomalies into three primary sources:

1. **Meter Malfunctions:** Trips recording positive fares with zero distance (1.40M instances).
2. **Financial Reversals:** Negative fares resulting from driver disputes or system refunds (2.85M instances).
3. **Logging Latency:** System timestamps where dropoff precedes pickup (0.55M instances).

9 Robustness / Sensitivity / Shift Analysis

The key temporal shift in 2025 stems from the introduction of the `cbd_congestion_fee`. Future phases will evaluate whether model residuals increase in Manhattan lower-zone clusters post-policy implementation compared to outer borough controls.

10 Ethics, Privacy and Governance

The dataset contains no Personally Identifiable Information (PII). Vendor IDs are coarse integer codes, and spatial locations are masked at the Taxi Zone level (265 spatial polygons), preventing individual driver or passenger tracking.

11 Discussion

The audit confirms that DuckDB is an optimal analytical engine for this project scale, performing full-year joins and aggregations in under 15 seconds on a standard 16GB laptop.

12 Limitations

External environmental factors (e.g., hourly rainfall, snowfall, subway service disruptions) are not present in the native TLC dataset and will require external data ingestion in Phase 2.

13 Conclusion

Phase 1 successfully establishes the analytical pipeline, quantifies data quality boundaries, and formalizes the prediction task. Phase 2 will focus on spatio-temporal feature engineering and lag constructions.

14 References

1. NYC Taxi and Limousine Commission, *TLC Trip Record Data (2025)*, official NYC Open Data portal.
2. DuckDB Development Team, *DuckDB: An In-Process Analytical Database Management System*, 2026.

15 Supplementary Material

Data audit script executable via `src/data_structure_audit.py` *and notebook* `notebooks/data_audit.ipynb`.